

Renewable energy technology diffusion: an analysis of photovoltaic-system support schemes in Medellín, Colombia



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ABSTRACT

Colombia's electricity mix is dominated by hydropower, which constitutes over 65 percent of the installed capacity. The country has important potential for introducing solar photovoltaic sources into its electricity generation mix given its high average annual insolation. However, there is a lack of incentives and support schemes for alternative renewable energy technologies. This paper analyzes the diffusion of a photovoltaic system in Colombia with a focus on Medellín, Colombia's second largest city. A diffusion model is constructed, based on the classic Bass diffusion theory, where the adoption rate is a function of awareness-raising campaigning and social interaction. The model incorporates both subsidy and feed-in-tariff policies. Policy implementation scenarios and the effects of policy mixes are analyzed. Results show that a 50 percent subsidy for investment, together with a USD 0.30/kWh feed-in-tariff rate, jointly provide the highest marginal increase in diffusion rate. However, given the cheapness of the country's current hydroelectric resources, photovoltaic-system diffusion might remain a challenge for both the government and private sectors for the foreseeable future.

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1. Introduction

Article 2 of the Kyoto Protocol framework aims for the implementation of policies for research and development of renewable energy (RE) resources, carbon sequestration technologies, and innovative environmentally-friendly technologies (UN, 1998). The 2002 World Summit on Sustainable Development at Johannesburg sought to review the achievements of the decade-old and laid down plans for implementation. Although no concrete targets for the increased consumption of RE resources have been established (WHO, 2013), such initiatives have influenced many countries' perspectives on the development of RE (ECLAC, 2004). At the summit's culmination, Colombia joined another 81 countries in establishing the Johannesburg Renewable Energy Coalition (JREC) that aimed to “focus on international, regional, and national political initiatives that will help foster policies for the promotion of renewable energy” (Europa, 2013). Three years after the Johannesburg Summit,

55 countries had adopted one or more RE policies. Six years later, this figure had grown to almost 120 (REN21, 2011). A variety of RE policies have been implemented in Colombia since the 1992 Rio Conference (IRENA, 2012). In parallel to the Colombian government's interest in RE, this paper investigates the diffusion of PV system, particularly in Medellín, the country's second largest city.

Colombia is located in the northern region of South America, bordering the Caribbean Sea, between Panama and Venezuela, and bordering the North Pacific Ocean, between Ecuador and Panama. Its total area is 1,138,900 km² with a tropical climate along the coast and eastern plains, while lower temperatures are experienced in the highlands (DANE, 2013). Population estimates in 2012 for Medellín were 2.743 million for the city and 3.590 million for the metropolitan area, with an annual population growth rate of 1.13 percent (CIA, 2013). In 2005, Medellín city had 612,115 households; in 2012, the number of inhabitants was estimated to be 783,000 (DANE, 2013). Hydropower is Colombia's major source of electricity generation, making up over 67 percent of its total electricity production in 2009 (Loy and Gaube, 2002; UPME, 2013). The relative proportion of each source has not changed significantly since the early 1990s (UPME, 2013). In 2009, 41.7 TWh (72.8 percent) of Colombia's 57.3 TWh of generated electricity, came from renewable resources (IRENA, 2012).

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In 2001, the Colombian Congress signed the *Ley 697 de 2001* (Law 697 of 2001) that promotes an efficient use of energy and the use of alternative energy sources (MinMinas, 2003). Thereafter, subsequent laws focused on targeting goals, tax exemptions, investment in research, and reliability-charge exemptions for small projects (IADB, 2010). Such initiatives were not effective in the deployment of RE resources because of the lack of support schemes (Ruiz-Mendoza and Sheinbaum-Pardo, 2010). For instance, Resolution 181,401/2004 (MinMinas, 2004) sets a target for CO₂ emission reduction through the use of non-conventional energy resources; however, no specific guidelines are mentioned regarding the incentives for installing a particular RE technology (RET), the kind of support a particular government agency will provide, or the timeline to reach the target. Existing policies are mostly limited to biofuels, CO₂ emission reduction, and encouragement of energy efficiency. Though overall targets are set, the legislation does not specify how the goals are to be reached.

The Colombian power system was restructured in the mid-1990s, moving from a government monopoly to a market-oriented structure (IADB, 2010), which also drew the attention of scholars interested in electricity deregulation. Thus, several studies have focused on understanding post-deregulation scenarios (Arango et al., 2006; Arango, 2007; Larsen et al., 2004); while others have centered on the adoption of RET in the liberalized market. For instance, Zuluaga and Dyer (2007) constructed a simulation model to analyze different policies that may encourage the adoption of RE in Colombia. Though they explored two alternative technology diffusion scenarios, they aggregated all RE resources into a cumulative installed capacity. Botero et al. (2009) used MARKAL (Market Allocation Model) to identify the optimal contribution of energy mixes to a cost minimization objective function. Lund et al. (2014) investigated how energy system in cities could be used as a carbon reduction strategy. The first grid-connected building featuring a photovoltaic system installation in Colombia was studied by Aristizábal and Gordillo (2008); but their experiment was limited to the technical feasibility, looking into the array, the inverter, and the system performance of the PV system.

South America is the region with the highest share of electrical production from renewable resources worldwide (Enerdata, 2012; Observ'ER, 2012). When it comes to solar energy, Chile and Brazil have the largest installations, with 51 and 37 percent, respectively, of the total PV capacity in the region (Solarbuzz, 2013). Chile's dominance in the South American PV market can be explained partly by effective energy policy (Choudhury, 2013; MinEnergia, 2012) rather than by the potential of PV power. Varieties of RE policies have been implemented already or are planned for Latin America, such as:

- **Ecuador:** The national electricity board, CONELEC, implemented a feed-in tariff (FiT) policy in April 2011. The program includes tariffs for wind, biomass, biogas, geothermal and hydroelectric plants up to 50 MW in size (Choudhury, 2013; CONELEC, 2011).
- **Argentina:** The government's target is to reach an eight percent share of its energy mix with renewable resources by 2016. It has a FiT system for solar power implemented through a national fund for the promotion of RE (Choudhury, 2013).
- **Mexico:** A renewable energy bill aims to reach a 35 percent target for use of RE resources by 2024. Tax incentives for solar projects and a net metering system have been implemented (Bisseger, 2013; Choudhury, 2013).
- **Brazil:** PV projects of ≤ 1 MW are being planned in preparation for the 2016 Olympics. Its electricity regulatory agency, ANEEL, is preparing to implement two new solar incentive policies: an 80 percent reduction in taxes and the implementation of net

metering for residential and commercial installations (Bisseger, 2013; Choudhury, 2013).

- **Chile:** There are 3.9 MW of PV projects installed and another 30 MW currently under construction (Lacey, 2013). Act 20,257 requires electrical utilities with an installed capacity of over 200 MW to obtain part of the energy they sell to customers from RE resources (Choudhury, 2013). A law on net metering is currently in Congress pending approval (MinEnergia, 2012).

Although the global average of RE production cost has drastically decreased in the past decade, developing countries' adoption of the technology vary significantly (Huenteler et al., 2014; Jänicke, 2012). Table 1 shows Colombia's potential for harnessing a wide range of RE (ECLAC, 2004). While countries that experience four distinct seasons have electricity consumption peaks during the coldest and warmest months of the year (Farmington, 2013; StatCanada, 2010), tropical countries like Colombia have a relatively constant temperature and seasonal pattern and, therefore, have more stable energy consumption throughout the year. With solar energy, the average radiation is 4.5 kWh/m²/day and the best solar resource area is the Guajira Peninsula, with insolation of 6 kWh/m²/day (UPME, 2002). In contrast, while Germany has the largest installed PV capacity in the world, at 24.8 GW as of 2011, it produces only around 3.8 kWh/m²/day at its peak, in Kempton in the Allgäu region, which has already the highest solar potential in the country (European Commission, 2008; Meteonom, 2007; Solarplaza, 2012).

Solar irradiance (W/m²; and, relatedly: insolation, kWh/m²/day) is a measure of how much solar power a location receives. It varies throughout the day according to the sun's position and throughout the year depending on the season (Boxwell, 2013). On average, Medellín receives 4.57 kWh/m²/day insolation (Table 2).

The steady decline of installed PV system costs in past decades is gradually making PV competitive in regions with high solar insolation and/or high electricity prices (IEA PVPS, 2012). Having the same advantage with other countries near the equator, Colombia receives a relatively stable amount of solar radiation throughout the year (Boxwell, 2013), providing it with high potential for solar energy. However, PV installations in Colombia remain very limited and mostly for research or experimental purposes only (Aristizábal and Gordillo, 2008; Loy and Gaube, 2002).

The main goal of this paper is to investigate the diffusion of renewable energy technologies in Colombia. Specifically, the effects of subsidy and FiT support-schemes on the deployment of PV system, with focus on the city of Medellín, are analyzed. Section 2 reviews the two support schemes included in this paper and discusses the Bass diffusion model and PV learning curve. In Section 3, the scope, assumptions, and model validation are discussed. Next, a comparison is made between the effects of different policy mixes in Section 4, including breakeven and sensitivity analyses. Finally, in Section 5, different scenarios are shown in support of the schemes. Recommendations on future research are also provided.

Table 1
Renewable energy potentials for Colombia (ECLAC, 2004).

Energy	Potential
Solar	Annual potentials from 5 to 6 kWh/m ² /day
Biomass	Annual production of cane <i>bagasse</i> is 7.5 M tons; rice husks at 457,000 tons
Wind	Over 10 m/s at the northern region
Hydro	50 GW of >100 MW capacity; 70 GW for medium and small scale
Geothermal	Greatest potential in Nariño, Los Nevados National Park, and Paipa
Tidal	Around 500 MW along the coast
Wave	Up to 30 GW along the coast

Table 2
Average insolation for Medellín (Boxwell, 2013).

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Mean
Solar insolation (kWh/m ² /day)	4.79	4.69	4.51	4.16	4.40	4.65	4.97	4.80	4.43	4.40	4.49	4.51	4.57
Optimum angle (degrees from vertical)	68°	76°	84°	92°	100°	108°	100°	92°	84°	76°	68°	60°	84°

2. Literature review

Returns on investment, subsidies on conventional energy sources, and the failure to internalize environmental externalities are common barriers to diffusion (Osmani et al., 2013; Sovacool, 2009; van Alphen et al., 2008). Effective energy support policies are necessary to catalyze the diffusion process until RETs can compete with conventional sources (Keyuraphan et al., 2012). While PV systems are one of the fastest growing RETs (REN21, 2011), they also remain one of the most expensive energy sources to date (Haas et al., 2011). A variety of policies, such as quota requirements, FiT, tax credits, portfolio standards, pricing laws, production incentives, trading systems, etc. are used to encourage the diffusion process (Badcock and Lenzen, 2010; Kissel and Krauter, 2006; Pegels, 2010). The categorization of RE policies aids in understanding their functions and in designing policies (Zhai, 2013). This study focuses on market-based, supply-push strategies, namely: *subsidy* and *FiT* policies, which are two common support schemes.

2.1. Capital subsidy

Capital subsidies (or simply, subsidies) either keep prices for consumers below market level or keep prices for producers above market level (De Moor, 2001), where the government absorbs the difference between the actual investment cost or production cost and the price level that will stimulate potential adopters to become actual adopters of RET (Moosavian et al., 2013). The three categories of subsidies are: financial subsidies, research and development funding, and external costs of energy production (UIC, 2005). Subsidy is usually pegged at a fixed percentage of the total investment cost (Cansino et al., 2011).

2.2. Feed-in-tariff

FiT is a cost paid by the government to the electricity supplier for every unit of electricity produced. This cost, in turn, is passed on to the consumer through a premium in the retail electricity price (Keyuraphan et al., 2012; Solangi et al., 2011). When electricity consumers act also as producers (e.g. by installing a PV system and feeding their surplus production into the grid), they can benefit financially from receiving a FiT rate from the utility (or the government) aside from paying reduced electricity bills (Bertoldi et al., 2013). Doherty and O'Malley (2011) discuss three elements of FiT policy, namely: floor price, balancing payment, and technology difference. A FiT can be either a guaranteed fixed price, regardless of the quantity of electricity produced; or a premium rate, which incentivizes the market price of the electricity produced (Jenner et al., 2013; Toke, 2007). A FiT guarantees incentives to electricity generators and pushes down RET costs; however, it causes a surge in the electricity price until economies of scale bring the price down to lower levels again (Sovacool, 2010).

2.3. Renewable energy technology diffusion modeling

Originally formulated by Frank Bass in 1969, 'Bass diffusion' describes the dynamics of innovation adoption: from introduction

to progression, via the apex, and then the declining phase of the diffusion process (Bass, 1969). The Bass model has been used for a wide variety of real-world problems such as forecasting (Chu and Pan, 2008; Tseng and Hu, 2009), new product growth (Cheng, 2012; Chiang and Wong, 2011; Kaldasch, 2011), and innovation diffusion (Cho and Koo, 2012; Meade and Islam, 2006; Peres et al., 2010).

Previous studies in RET diffusion include PV system adoption in Tennessee's poultry industry (Bazen and Brown, 2009), India (Peter et al., 2006), and the United States and Japan under the lenses of an innovation value-added chain framework (Shum and Watanabe, 2009). In addition, RET diffusion studies based on Bass diffusion include forecasting cross-country PV adoption patterns (Guidolin and Mortarino, 2010), promotion policies for PV and solar water-heater systems in Japan (Yamaguchi et al., 2013), as well as diffusion policy issues in general (Rao and Kishore, 2010). Hence, both general and Bass RET diffusion analyses are well-established and widely used. Thus, they serve as the starting point for the analysis of PV diffusion in the city of Medellín, Colombia.

Two main agents of diffusion dynamics are the innovators and imitators. Innovators take information from external sources such as campaigns or advertisements, while imitators are those that source information through social interactions (Bass et al., 1994). Mathematically, the Bass diffusion is expressed (Guidolin and Mortarino, 2010) as:

$$z'(t) = p \times (m - z(t)) + q \times (z(t)/m) \times (m - z(t)) \quad (1)$$

where: $z'(t)$ = adoption rate; m = potential market; $z(t)$ = potential adopters; thus $(m - z(t))$ = remaining market; p = campaign effectiveness; q = contact rate; and t = time. With the closed-population assumption, potential adopters decrease as more people become actual adopters. This implies that the component of $z'(t)$ resulting from campaign, $p \times (m - z(t))$, eventually decreases over time as the potential market becomes saturated with knowledge about the production (i.e. PV system). Social interaction, $q \times (z(t)/m) \times (m - z(t))$, on the other hand, results from the contact of the potential adopters with the whole population. That is, there is no distinction between potential and actual adopters. So $z'(t)$ reaches its maximum value when the actual adopters and $z(t)$ are equal. The diffusion model in the current study aims to determine the influence of policy mixes on the actual adopters (A_a), expressed as:

$$A_a = \sum (WH_t + WA_p).$$

Factoring out W ,

$$A_a = W \times \sum (H_t + A_p), \quad (2)$$

where: A_a = actual adopters; A_p = potential adopters; H_t = total households; and W = willingness to invest in PV. As soon as potential adopters become actual adopters, they automatically install the PV system. Therefore, the PV diffusion module and the installed PV capacity are co-flows. That is, construction and adoption rates flow at the same speed. Average construction time is assumed to be 1 week (or 1/52 year). Hence, after a week, it is expected that the PV

capacity under construction will become installed PV capacity (PV_{ins}), expressed as:

$$PV_{ins} = \sum \left(\left(z'(t) \times C_i/L \right) - d \right) \quad (3)$$

where: C_i = construction rate; d = depreciation rate; and L = average lifetime of the PV system. The first monomial, $z'(t) \times C_i/L$, is the construction rate (CR). C_i increases when: (1) CR increases and d remains constant; (2) CR increases more than the rate of increase of d ; and (3) CR remains constant while d decreases. Also, d depends on the current value of C_i . When PV_{ins} increases or decreases, d also increases or decreases proportionally. When subsidy is implemented, the government shoulders a percentage of the investor's (i.e. household's) total investment cost. PV subsidized cost (PV_{sc}) is computed as:

$$PV_{sc} = PV_{ec} \times (1 - s)/100 \quad (4)$$

where: PV_{ec} = effective PV cost, and s = percentage of subsidy by the government. PV_{ec} is the total cost of installing the PV system, including PV module, balance-of-system (BOS), and the labor. Therefore, PV_{sc} is the residual cost that the actual adopter needs to spend to install the PV system. Another key variable in the diffusion model is the breakeven percentage (BE_p). It refers to the percentage of the conventional electricity price at which the PV cost competes. It is computed as follows:

$$BE_p = 1 - ((PV_{ac} - E_p)/PV_{ac}) \quad (5)$$

where: E_p = conventional electricity price and PV_{ac} = PV average cost. A value of 100 percent means parity between both costs; less than 100 percent means the PV cost is not competitive, while greater than 100 percent means cost competitiveness.

2.4. PV learning curve

A learning curve estimates the rate of decrease in production costs that can be attributed to the efficiency gained with cumulative production experience (Barreto and Kypreos, 2004; Desroches et al., 2013; Papineau, 2006). Cumulative production experience drives down PV costs as cumulative installed PV capacity increases. It is an important source of non-linearity in the behavior of a system. The general equation of a learning curve is (Argote and Epple, 1990):

$$C_t = C_0 \times (Q_t/Q_0)^b \quad (6)$$

where: C_t = cumulative production unit cost at time t ; C_0 = unit cost at t_0 ; Q_t = cumulative production at time t ; Q_0 = cumulative production at t_0 ; and b = learning index. The value of b becomes negative with increasing production levels. The residual percentage of the cost is referred to as the progress ratio (Argote and Epple, 1990; Van Sark, 2012). Progress ratios on PV modules are similar on global and local scales because modules are made by companies operating internationally (Bhandari and Stadler, 2009). In this paper, a progress ratio of 80 percent is considered (De La Tour, Glachant and Mènière, 2013; Junginger et al., 2005; Zhai, 2013).

3. Model construction

In this paper, the diffusion model seeks to show the causalities between the diffusion process, the effect of cumulative installed PV capacity on installation costs, and the willingness of the potential market to adopt the technology. As shown in Fig. 1, the campaign effectiveness depends on the remaining potential adopters, where

a proportional relationship between the two variables exists and may include strategic and tactical decisions (Guiltinan, 1999). The strength of campaign is a function of the growth of the total number of households over time; that is, the increasing number of people who share their experiences with others will have a positive overall effect on the campaign. Both PV capacity under construction and installed PV capacity increase proportionally with increasing numbers of adopters. However, the learning curve pushes the cost of production down as the cumulative quantity of production increases over time (Neij, 1997; Papineau, 2006). As effective PV cost decreases, both payback period and average annual cost decrease. These reinforce the feedback to the perceived investment willingness, which in turn affects both campaign and social interaction.

3.1. Scope and assumptions

The diffusion model covers a 20-year time horizon, from 2015 to 2035. Historical data are from 2000 to 2014 (15 years), while policies are implemented from 2015 to 2035 (20 years). Demographic factors such as births, deaths, and immigration (movement of family within or from outside the metropolitan area) are excluded. Each household has one family. All households are considered potential adopters. The average conventional electricity price from 2009 to 2013 (0.13, 0.13, 0.14, 0.16, and 0.16, respectively, in USD/kWh) is used in the model, based on actual average market price (UPME, 2013). There are four kinds of PV technologies currently available in the market: mono-crystalline, poly-crystalline, amorphous, and hybrid (Evo Energy, 2013), each having its advantages and disadvantages in terms of performance, efficiency, reliability, cost, and flexibility (Horizon Renewables, 2013; PV Resources, 2013; SolarFacts, 2013). In the model, no distinction between the types of technology is made. Annual PV electricity production of 4.5 GWh is considered. Detailed computations on the module price, BOS, labor cost, and maintenance are excluded. Average PV installed cost of USD 3500 is used (Greentech Media, 2013; Haas et al., 2013). Maintenance cost is assumed to be 5 percent of the total PV cost. An average PV system lifetime of 25 years is assumed in the model (PV Resources, 2013). PV system capacity varies according to the preferences of the household. In the model, a 4.5 kilowatt-peak (kWp) is considered.

3.2. Model validation

Validation is the process of building confidence in the model. The model was tested under five complexity validation levels, namely: elementary level, simple dynamics, multiple dynamics, full dynamics, and meta-level (Groesser and Schwabinger, 2012). The standard structural and behavioral tests (Barlas, 1989, 1996; Qudrat-Ullah and Seong, 2010) were also used to check the validity of the model. The following findings validate the robustness of the model: When the Feed in Tariff is reduced to zero, error results because zero *total revenue* as the denominator of *payback time* will result in an infinitely large value. When *average PV capacity* is set to zero, error results because *average annual PV cost* will be zero and *PV to conventional electricity cost index* will result in an infinitely large value. When *investment perception delay*, which has a default value of 3 years, is replaced by a very large number (e.g. 100), the *potential adopters* in the diffusion module do not become *actual adopters*. Consequently, *construction rate* and *installed PV capacity* remain zero. This makes sense because the more people in the potential market delay their investment decision the more actual installations of the PV system will be delayed. Since the time horizon is only up to year 2035, no construction will be observed during this period. When *average installed cost* is significantly large (e.g. USD 1,000,000), no *potential adopters* become interested in the

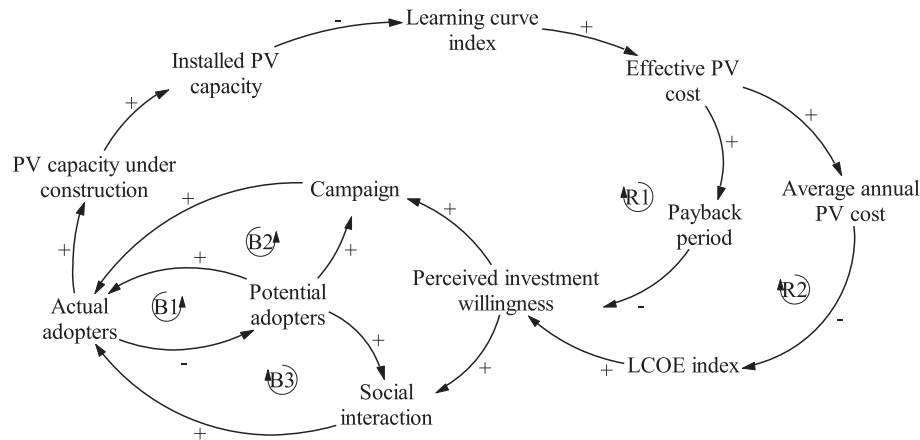


Fig. 1. Causal-loop diagram.

PV system, thus *adoption rate*, *actual adopters*, *construction rate*, and *installed PV capacity* remain zero.

3.3. Structure-behavior analysis

The structure of a system induces behavior over time (Meadows, 2008). Such relationships between structure and behavior can be analyzed with the phase diagram (also called phase plot, phase space, or structure-behavior diagram) (Stermann, 2000). A phase diagram plots two variables on a two-dimensional graph (Herbst et al., 2013) with independent and dependent variables plotted on the X and Y axes, respectively. Since understanding structure-behavior associations is an important prerequisite for designing policies (Davidsen, 1992), the dynamic relationships were explored between key variables in the model and their policy implications were analyzed. Fig. 2 plots PV *subsidized cost* (x-axis) versus *actual adopters* (y-axis). *Subsidized cost* is the PV installed cost less the subsidy from the government. The slightly convex curve formed by the dots from the upper left to the lower right part of the graph implies that a decrease in the PV *subsidized cost* results in a non-linear increase in *actual adopters*. That is, for every unit decrease in the PV *subsidized cost*, there is more than one unit increase in *actual adopters*. This non-linear relationship shows that when the perceived financial burden decreases, the potential market increases. Fig. 2 also shows the phase diagram of *annual policy budget* (x-axis) versus *actual adopters* (y-axis). The plot is segmented into two main regions marked on the graph: the first with an

exponential increase and the second without any evident relationship. The first part (*Region 1* in the graph) shows that increasing the *annual policy budget* results in an exponential increase in *actual adopters*. In the second part (*Region 2* in the graph), the vertical segment of the curve means that an increase in the *annual policy budget* is no longer needed to increase further the *actual adopters*.

4. Results and discussion

Two types of policies are considered in this paper: subsidy and FiT. Subsidy occurs when the government shoulders a fraction of the electricity producer's investment spending (De Moor, 2001); while FiT is the price paid by the government for every unit of electricity produced (Keyuraphan et al., 2012).

4.1. Policy scenarios

The policy-mix tree in Fig. 3 classifies different policies or mechanisms. On the first level (yellow (in the web version)), there are four scenarios, namely: *Status quo* (Baseline), *Kickoff*, *Subsistence*, and *Comprehensive*. Under the *Comprehensive* scenario, two schemes (2nd level; purple) are available: *Baseline* and *Intermediate*. Furthermore, for the *Intermediate* scheme, there are six options (3rd level; green) to choose from.

Under the *Status quo* scenario, no policy is implemented. When the policy switch is off, the model is in equilibrium. Implementation of neither policy results in zero both actual adopters and

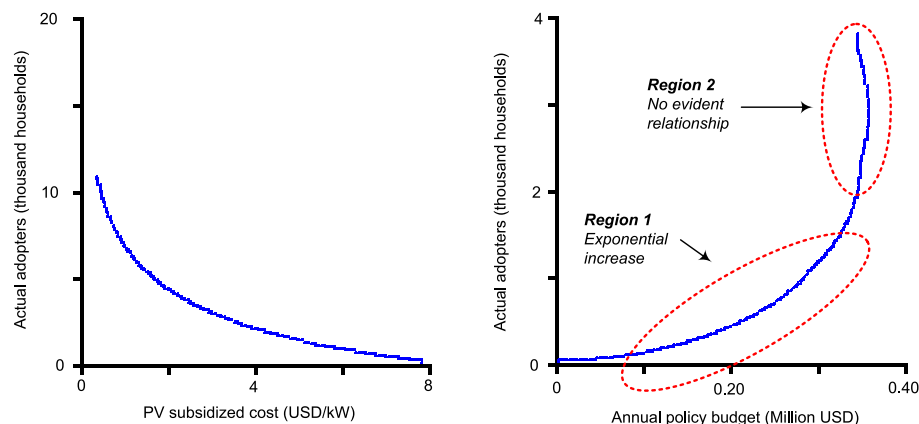


Fig. 2. Left: Phase diagrams of PV subsidized cost (x-axis) vs. actual adopters (y-axis). Right: Phase diagram of annual policy budget (x-axis) vs. actual adopters (y-axis).

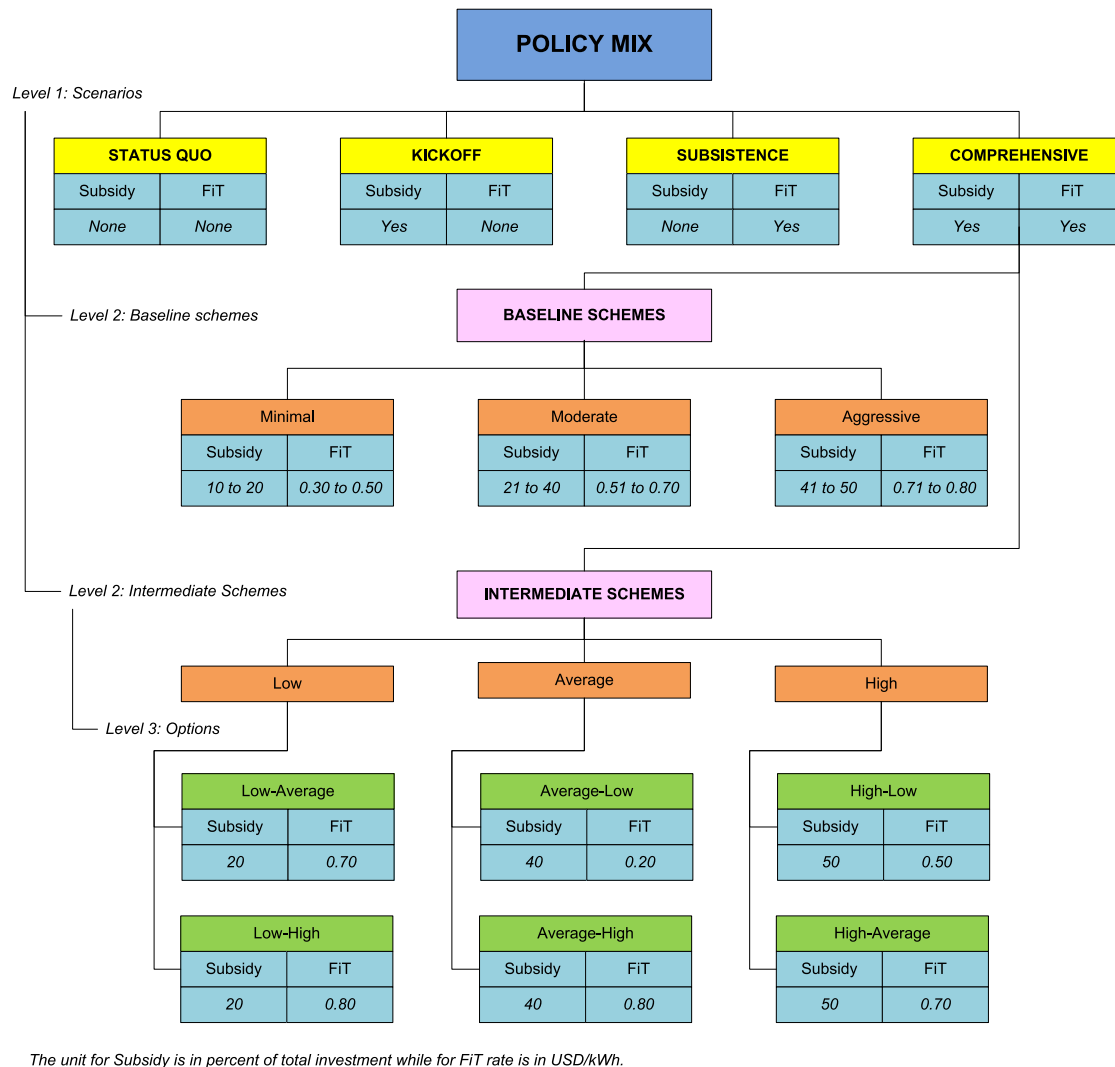


Fig. 3. Policy mix tree.

installed PV capacity. Under the *Kickoff* scenario, a subsidy is implemented, which works such that the government shoulders a fraction of the total investment cost of the household that wants to install a PV system without the implementation of the FiT. The *Kickoff* scenario shows no signs of a diffusion process taking place, regardless of the value of the subsidy. Even when the subsidy is at 50 percent (i.e. only half of the total investment in a PV system is spent by the household), no households become actual adopters. Under the *Subsistence* scenario, only the FiT is implemented. The minimum value of FiT that starts diffusion is USD 0.30/kWh, with only 12 actual adopters and 43 kW installed capacity by year 2035. An increase in the FiT rate up to USD 0.40/kWh increases the total installed PV capacity up to 847 kW by year 2035. When the FiT rate is USD 0.80/kWh, actual adopters increase to 902, while the installed PV capacity is 3.16 MW.

Under the *Comprehensive* scenario, both subsidy and FiT are implemented. The purpose of establishing the *Baseline* scheme is to divide the range of values for subsidy and FiT into segments. These ranges are separated into three categories, namely: *Minimal*, *Moderate*, and *Aggressive*. In addition to the three baseline schemes, six *Intermediate* schemes are presented. Comprehensive scenario results are presented in Table 3. A counterintuitive finding is that lower PV subsidized costs do not necessarily mean higher diffusion rates. For instance, compare Cases A and C. PV subsidized cost in

Case A is USD 12,595; while in Case C it is USD 9450. Although in Case A the PV subsidized cost is only USD 3145 higher than in Case C, the former has 904 more actual adopters than the latter.

4.2. Breakeven analysis

Breakeven analysis is an important economic decision-making instrument used to determine and analyze the point at which revenues equal total costs (Lesure, 1983; Powers, 1987). In RE parlance, this refers to *grid parity*, which is the point where the price of solar energy is equal to that of the conventional electricity source (IRENA, 2013; Yang, 2010). However, grid parity is excluded in the current study since *Levelized Cost of Electricity (LCOE)* is used in the computation instead of *average cost* (Branker et al., 2011; Hernández-Moro and Martínez-Duart, 2013). The sixth column of Table 3 shows the breakeven percentages (BE_p) of the nine Comprehensive scenarios at year 2035. BE_p is based on the conventional electricity price and the PV average cost as shown in Eq. (5). In the case of the *Minimal Baseline* scheme, for instance, the PV average cost would be 16 percent of the conventional electricity price at year 2035.

Breakeven analysis has policy implications. It assists in determining how soon the PV cost can compete with the conventional electricity price. For instance, the *Aggressive* scheme breaks even at

Table 3
Baseline scenarios and policies.

Scheme	Description	Policy mix (subsidy, FiT rate) in (percent, USD/kWh)	Actual adopters by year 2035 (households)	PV installed capacity (MW) by year 2035	Breakeven percentage (%) by year 2035	Breakeven year	Policy implementation fund (million USD) from 2015 to 2035	Funding gap (million USD) at year 2015
Baseline	Minimal	(10, 0.30)	70	0.26	16	—	—	—
	Moderate	(40, 0.70)	1272	4.47	80	2039	8.681	4.363
	Aggressive	(50, 0.80)	1657	5.82	120	2034	9.019	4.533
Inter-mediate	A (low, moderate)	(20, 0.70)	977	3.42	56	2045	2.600	1.307
	B (low, high)	(20, 0.80)	1106	3.88	67	2042	2.874	1.445
	C (moderate, low)	(40, 0.20)	73	0.25	15	—	—	—
	D (moderate, high)	(40, 0.80)	1415	4.97	95	2036	6.972	3.504
	E (high, low)	(50, 0.50)	1104	3.87	64	2042	7.175	3.606
	F (high, moderate)	(50, 0.70)	1470	5.17	99	2036	8.966	4.506

year 2034. Starting that year, the cost of conventional electricity is already equal to or higher than the PV cost. When this happens, it would be more cost effective for the consumer to use electricity from PV than from conventional sources. From the government's perspective, assessing the breakeven year has potential for significant cost reductions. As soon as the breakeven year is reached, it becomes less imperative for the government to provide further incentives for the potential market to install PV systems since the government's goal on diffusion rate has already been reached. The government may start to amend the policy mix, usually decreasing either the subsidy or the FiT, or both, to help reduce the annual policy budget while sustaining PV diffusion targets.

4.3. Policy mix cost implications

Cost-benefit analysis (CBA) compares policy options (Courard-Hauri, 2004) by measuring their financial impacts, total net benefits, net present value of benefits and costs (Ekren et al., 2009), and rejecting the provision of funds to options that do not provide sufficient benefits or which do not pass certain criteria (Hof et al., 2008; Wegner and Pascual, 2011). While CBA assists in optimally allocating resources (Doeleman, 1985), several issues such as uncertainty, risk, and setting of boundary conditions remain as challenges in the decision-making process (Maciariello, 1975). In this paper, CBA is carried out by determining the *policy implementation fund* (F_{pi}). F_{pi} is expressed as:

$$F_{pi} = \sum (PV_{ec} - PV_{sc}) \quad (7)$$

where: PV_{ec} = PV effective cost and PV_{sc} = PV subsidized cost. F_{pi} is the total budget requirement for the entire policy implementation period (Table 3). The net present value (NPV) (Ekren et al., 2009) of the F_{pi} over the 20-year implementation period (from 2015 to 2035) is the estimated budget that the government needs to allocate at the start of the implementation. This NPV is referred to as the *funding gap* (F_g), which is the amount of investment needed at the start of the implementation period to ensure the successful implementation of a project. By assuming a 3.5 percent annual interest rate (Green Book, 2011), this budget should be sufficient to cover PV installation costs for all actual adopters according to the subsidy rate of the policy. F_g is expressed as:

$$F_g = F_{pi} / (1 + r)^t \quad (8)$$

where: r = discount rate – 3.5 percent is considered (Green Book, 2011) – and t = 20 years. Before implementation, financial implications of policies should be considered. F_g may be a decisive factor for the government because fiscal planning is on an annual basis. This means that in order for the policy to be effective during the entire implementation period, the government should be willing to

allocate this budget at the first year. Table 3 shows the trade-off between BE_p and F_g . The *Aggressive* scheme has the largest diffusion rate (A_d) and BE_p . However, it also has the largest F_g . At the other extreme, the *Intermediate* scheme, *Case A*, has the lowest A_d and BE_p , but also the lowest F_g . The most aggressive policy mix has the largest cost implications.

In principle, RET diffusion's goal is to maximize actual adopters. Hence, the *Aggressive* scheme would be the most favorable option. In some cases, constraints (e.g. availability of funds) may impede the efficacy of the actual implementation, thus settling into sub-optimal strategies. Trade-offs between policy effectiveness and implementation costs exist. Therefore, weighing in factors such as risk preferences and priorities becomes apparent. Policy mix should be considered not as a 0-1 or black-and-white option, but as a continuum of strategies. Similar to a management flight simulator (Stermann, 2000) where the decision maker is able to change inputs to obtain desired results, a policy mix should be able to be modified to fit under various constraints.

4.4. Sensitivity analysis

Sensitivity analysis examines the interaction between model behavior and variations in input parameters (EC, 2000; Saltelli et al., 1993; Tan and Tian, 2013). It informs decision makers about the reliability of the results through simulation (Borgonovo and Tarantola, 2012). In this paper, the sensitivity analysis is used to analyze the impacts of changing the subsidy and FiT on key diffusion variables. To analyze the sensitivity of the model behavior to subsidy, the FiT rate is held constant at its default value (USD 0.40/kWh). Every 10 percent increase in subsidy corresponds to a constant (USD 1800) decrease in PV subsidized cost. This linear relationship, however, does not hold true for actual adopters. In Table 4, a 10 percent increase in subsidy leads to a non-linear increase in actual adopters. A 10 percent subsidy increase from 40 to 50 percent leads to a 162-household increase in adopters, which has the highest difference among the 10 percent ranges. This implies that, holding other parameters constant, the 40 to 50 percent range provides the highest marginal increase in PV diffusion.

Table 4
Subsidy sensitivity analysis results.

Variable	Subsidy (percent)					
	0	10	20	30	40	50
PV subsidized cost (USD)	14,590	12,631	10,788	9072	7323	5715
Difference (USD)	—	1959	1843	1716	1749	1608
Actual adopters (household)	242	357	472	581	738	900
Difference (household)	—	115	115	109	157	162

Table 5
FiT sensitivity analysis results.

Variable	FiT rate (USD/kWh)							
	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80
BEP (percent)	7	13	22	32	43	54	66	79
Difference (percent)	—	6	9	10	11	11	12	13
Actual adopters (household)	0	0	315	581	803	959	1105	1251
Difference (household)	—	0	315	266	222	156	146	146

To analyze the sensitivity of the model to the FiT, the subsidy rate is held constant at 30 percent. Table 5 shows that a 0.10 USD/kWh marginal increase in FiT results to a 10 percent constant increase in BE_p . For actual adopters, however, a non-linear relationship exists. Results show that a FiT increase from 0.20 to 0.30 USD/kWh corresponds to the highest marginal increase in actual adopters. This implies that, holding other parameters constant, the 0.20 to 0.30 range provides the highest marginal increase to PV diffusion. When a 50 percent subsidy is combined with a USD 0.30/kWh FiT, A_a reaches 631 households, PV_{ins} is 2.21 MW, and BE_p is 33 percent in year 2035. Consequently, USD 2.246 million is required at the first year of implementation. From an economic perspective, determining the quantity of additional output for every unit of input is straightforward if a linear relationship exists. The analysis, however, quickly becomes confounding when two or more inputs are introduced together. The complexity is attributed not to the quantity of inputs, but to the dynamic responses among the inputs themselves. In the same manner, the effects of simultaneous implementation of subsidy and FiT need to be considered. This is of interest to understand the point of inflection of the combined outcomes of multiple policies.

5. Conclusions

Colombia's potential for photovoltaic electricity generation is characterized by constant daytime sunshine hours throughout the year, which is an advantage as compared with countries that experience four distinct seasons. One of the factors that hinder PV diffusion is cheap electricity from hydropower, mainly because Colombia has abundant water. Although the country has implemented over nine renewable energy policies since the year 2000, there is a lack of support schemes for non-conventional renewable electricity sources such as wind and PV. Unlike Ecuador and Argentina, where PV support schemes are currently in place, installation of PV systems in Colombia for residential consumers will remain a luxury in the foreseeable future, until concrete renewable energy technology policies are implemented.

A causal-loop diagram was constructed to show how five overlapping feedback loops influence the behavior of the diffusion model. *Adoption*, *Campaign*, and *Social Interaction* feedback loops directly affect the increase in actual adopters, while *Payback Period* and *Annual Cost* feedback loops affect the perceived willingness of the potential adopters to invest in PV system installation.

The diffusion model showed that an aggressive scheme (50 percent subsidy and 0.80 USD/kWh FiT) yields the highest diffusion rate (1657 households by year 2035). Consequently, this also has the most expensive policy implementation budget (USD 4.533 million) in the first year of implementation. An increase from 40 to 50 percent in subsidy and from 0.20 to 0.30 USD/kWh in FiT together provides the highest marginal increase in actual adopters. These ranges are important when assessing the minimum values of the policy mix that would yield the highest marginal increase in actual adopters.

In addition, the non-linear relationships between the variables in Fig. 2 also have policy implications. It is important for a decision

maker to understand that increased policy effectiveness does not necessarily require higher budget allocation. The existence of limitations on the effects of budget increases indicates that a policy mix can be effective even with a limited policy implementation budget. Decision makers (or policy makers) are interested to determine the most effective policy mix. Policy effectiveness can be evaluated from the standpoint of either objectives or budget constraints. Given the different policy mixes, the decision maker needs to understand the trade-off between costs and benefits. Is the decision maker (e.g. government) willing to allocate such an amount to clean energy rather than to other priorities, for instance, education, infrastructure, social welfare, or health? Does the government have enough funds to sustain the policy during the entire implementation period? Addressing these questions is a good starting point for discussion. Choosing the best policy mix or support scheme is a complex, and usually time-consuming, political process. It requires the consideration of other government priorities.

From a myopic point of view, it may be most reasonable to opt for policies that have short-term outcomes. However, causes and effects are distant in time and space and a careful consideration of a policy's long-term effects on the society as a whole is a decisive factor. Human comprehension is largely constrained to information available for decision-making. Simulation models bridge this gap by providing insights, sometimes counterintuitive ones, which improves the understanding of the system. They are able to compress time and space and allow to validate people's limited understanding by allowing them to play with decision variables that are otherwise difficult, expensive, or simply impossible to control in the real world.

Soft variables important for diffusion dynamics (e.g. campaigns, social interaction, payback period (a benchmark), and willingness) were included in the model. Results showed that their effects on model behavior are important. However, no data are available to precisely quantify or measure them. Group model building could be used in future research to improve soft-variable elicitation. Effects of policy mixes based on subsidy and FiT were presented. Exploring other fiscal and non-fiscal incentives may provide more strategic options in reaching diffusion targets. The RE learning curve plays important roles in RET costs and should be up-to-date.

In this paper, parameter values (e.g. PV capacity, BOS, maintenance cost, PV electrical production, PV lifetime) are based on global averages. *Investment willingness*, which is the expectation of potential adopters to become actual adopters, is assumed to drive campaign and social interaction. Future research could explore alternative methods to quantify or measure willingness more accurately. Future research may also focus on the economic status of the households since this variable has important implications for the potential adopter's willingness to invest in PV. While high-income households have more inclination to invest, low-income households would prefer whatever technology provides them with the lowest cost.

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